L. Yu, S. Wang, X. Li, C.-W. Fu, and P.-A. Heng, “Uncertainty-Aware Self-ensembling Model for Semi-supervised 3D Left Atrium Segmentation,” in *MICCAI 2019*, vol. 11765 , pp. 605–613.

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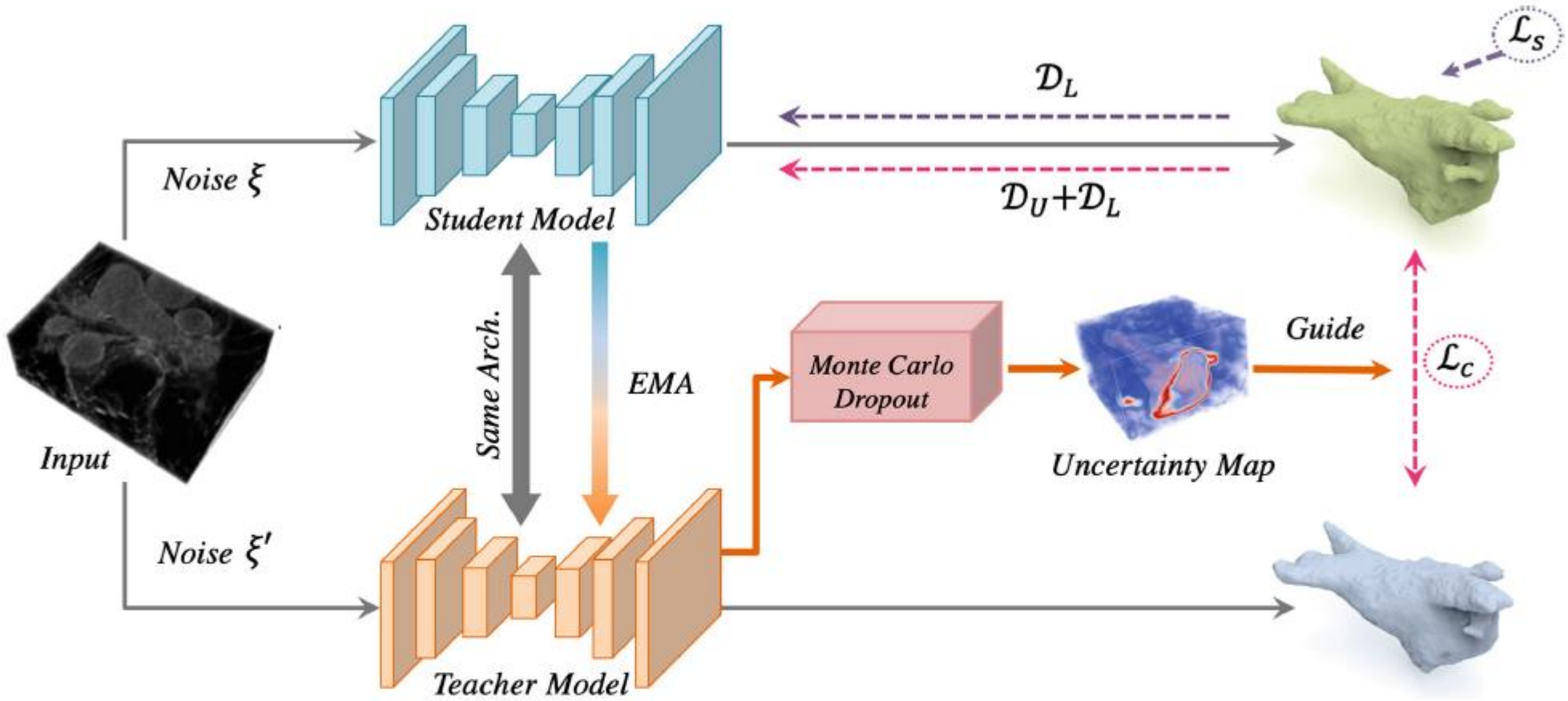

Fig. 1. The pipeline of our uncertainty-aware framework for semi-supervised segmentation. The student model is optimized by minimizing the supervised loss $\mathcal{L}_s$ on labeled data $\mathcal{D}_L$ and the consistency loss $\mathcal{L}_c$ on both unlabeled data $\mathcal{D}_U$ and labeled data $\mathcal{D}_L$. The estimated uncertainty from the teacher model guides the student to learn from the more reliable targets from the teacher.

Loss Function

We study the task of semi-supervised segmentation for 3D data, where the training set consists of N labeled data and M unlabeled data. We denote the labeled set as $\mathcal{D}_L = \{(x_i, y_i)\}_{i=1}^{N}$ and the unlabeled set as $\mathcal{D}_U = \{x_i\}_{i=N+1}^{N+M}$, where $x_i \in \mathbb{R}^{H \times W \times D}$ is the input volume and $y_i \in \{0,1\}^{H \times W \times D}$ is the ground-truth annotations. The goal of our semi-supervised segmentation framework is to minimize the following combined objective function:

$$\min_{\theta} \sum_{i=1}^{N} \mathcal{L}_s(f(x_i;\theta), y_i) + \lambda \sum_{i=1}^{N+M} \mathcal{L}_c(f(x_i;\theta',\xi'), f(x_i;\theta,\xi)), \tag{1}$$

Uncertainty Estimation

Uncertainty Estimation. Motivated by the uncertainty estimation in Bayesian networks, we estimate the uncertainty with the Monte Carlo Dropout [8]. In detail, we perform T stochastic forward passes on the teacher model under random dropout and input Gaussian noise for each input volume. Therefore, for each voxel in the input, we obtain a set of softmax probability vector: $\{\mathbf{p}_t\}_{t=1}^{T}$. We choose the predictive entropy as the metric to approximate the uncertainty, since it has a fixed range [8]. Formally, the predictive entropy can be summarized as:

$$\mu_c = \frac{1}{T}\sum_t \mathbf{p}_t^c \qquad \text{and} \qquad u = -\sum_c \mu_c \log \mu_c, \tag{2}$$

where $\mathbf{p}_t^c$ is the probability of the c-th class in the t-th time prediction. Note that the uncertainty is estimated in voxel level and the uncertainty of the whole volume U is $\{u\} \in \mathbb{R}^{H \times W \times D}$.

Uncertainty-Aware Consistency Loss

Uncertainty-Aware Consistency Loss. With the guidance of the estimated uncertainty U, we filter out the relatively unreliable (high uncertainty) predictions and select only the certain predictions as targets for the student model to learn from. In particular, for our semi-supervised segmentation task, we design the uncertainty-aware consistency loss $\mathcal{L}_c$ as the voxel-level mean squared error (MSE) loss of the teacher and student models only for the most certainty predictions:

$$\mathcal{L}_c(f', f) = \frac{\sum_v \mathbb{I}(u_v < H) \left\| f'_v - f_v \right\|^2}{\sum_v \mathbb{I}(u_v < H)}, \tag{3}$$

where $\mathbb{I}(\cdot)$ is the indicator function; f'_v and f_v are the predictions of teacher model and student model at the v-th voxel, respectively; u_v is the estimated

实验

Table 1. Comparison between our method and various methods.

Method	# scans used		Metrics			
	Labeled	Unlabeled	Dice [%]	Jaccard [%]	ASD [voxel]	95HD [voxel]
Vanilla V-Net	16	0	84.13	73.26	4.75	17.93
Bayesian V-Net	16	0	86.03	76.06	3.51	14.26
Vanilla V-Net	80	0	90.25	82.40	1.91	8.29
Bayesian V-Net	80	0	91.14	83.82	1.52	5.75
Self-training [1]	16	64	86.92	77.28	2.21	9.19
DAN [18]	16	64	87.52	78.29	2.42	9.01
ASDNet [12]	16	64	87.90	78.85	**2.08**	9.24
TCSE [10]	16	64	88.15	79.20	2.44	9.57
UA-MT-UN (ours)	16	64	88.83	80.13	3.12	10.04
UA-MT (ours)	16	64	**88.88**	**80.21**	2.26	**7.32**

Thank You!